



Cryptography

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Week 3

Module 1

- Symmetric Ciphers
- Block ciphers
- SPN
- Feistel Structure referring to DES

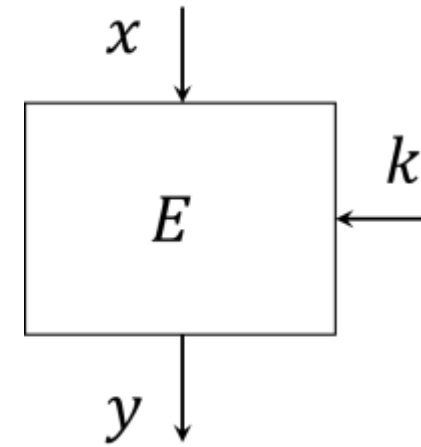
Symmetric cryptosystems

- The cryptosystems in which the decryption key can be derived efficiently from the knowledge of the encryption key are called symmetric cryptosystems.
- Symmetric cryptosystems are divided into stream ciphers and block ciphers.
- Presently, we will discuss block ciphers.



Block ciphers

- A block cipher is a cryptosystem that transforms an n -bit plaintext block to any n -bit ciphertext block using an m -bit key.



- A block cipher is represented by a function

$$E: \{0, 1\}^n \times \{0, 1\}^m \rightarrow \{0, 1\}^n$$

where $\mathcal{P} = \mathcal{C} = \{0, 1\}^n$ and $\mathcal{K} = \{0, 1\}^m$.

- The adjacent figure is a diagram of a block cipher.

Iterated Block Ciphers: the Key Schedule

- Let K be a random binary key of some specified length.
- K is used to construct NR round keys (also called subkeys), which are denoted by $K^1, \dots, K^{NR}$.
- The list of round keys $K^1, \dots, K^{NR}$ is called the key schedule.
- The key schedule is constructed from K by using a fixed public algorithm.

Iterated Block Ciphers: the round function



- The round function, say g , takes two inputs: a round key (K^r) and a current state denoted by w^{r-1} .
- The initial state w^0 is defined to be the plaintext, x .
- The ciphertext y is defined to be the state after all NR rounds.

The round function: encryption

- The encryption operation is carried out as follows:

$$\begin{aligned}w^0 &\leftarrow x \\w^1 &\leftarrow g(w^0, \kappa^1) \\w^2 &\leftarrow g(w^1, \kappa^2) \\&\vdots \\w^{NR-1} &\leftarrow g(w^{NR-2}, \kappa^{NR-1}) \\w^{NR} &\leftarrow g(w^{NR-1}, \kappa^{NR}) \\y &\leftarrow w^{NR}\end{aligned}$$

The round function: decryption

- For decryption to be possible the function g must have the property that it is injective (one-to-one) if its second argument is fixed.

$$\begin{aligned}w^{NR} &\leftarrow y \\w^{NR-1} &\leftarrow g^{-1}(w^{NR}, \kappa^{NR}) \\&\vdots \\w^1 &\leftarrow g^{-1}(w^2, \kappa^2) \\w^0 &\leftarrow g^{-1}(w^1, \kappa^1) \\x &\leftarrow w^0\end{aligned}$$

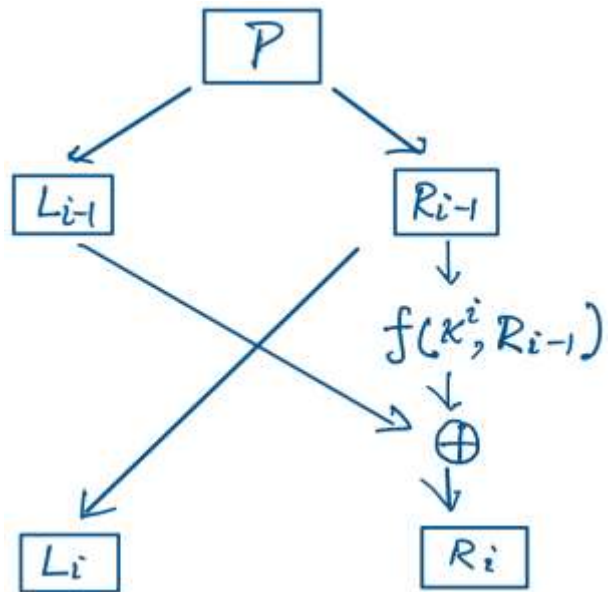
Feistel Cryptosystem

- A Feistel cryptosystem (also known as a Feistel network) is a block cryptosystem determined by the following components:
 - The block size $2t$ (an even number) and a key size N
 - The number of rounds NR (a positive integer)
 - A key schedule: A mechanism for generating NR round keys $\kappa^1, \kappa^2, \dots, \kappa^{NR}$ from the single cryptosystem key κ .
 - A round key function, f_{κ^i} , for each round key κ^i , which inputs any t -bit string R , and outputs another t -bit string $f_{\kappa^i}(R)$.

Encryption and Decryption

- Encryption
 - Given a plaintext P , a bit string of length $2t$, we first split P into two bit strings of length t : $P = (L_0, R_0)$, with L_0 being the left half, and R_0 being the right half.
 - We then proceed through NR rounds for the following transformations:
For round $i = 1$ to NR : $L_i = R_{i-1}$, $R_i = L_{i-1} \oplus f_{\kappa^i}(R_{i-1})$
 - The ciphertext will then be $C = (R_{NR}, L_{NR})$
- Decryption
 - We feed the ciphertext C back into the above encryption process, the only change being that the keys are used in the reverse order.

Feistel network



A simple three round Feistel cryptosystem

- The block size = 8. The key length = 12. The number of rounds = 3.
- The key scheduling algorithm
 - For each length-12 bit string $\kappa = k_1 k_2 \dots k_{12}$ the round keys are

$$\kappa^1 = k_1 \dots k_4 \oplus k_5 \dots k_8$$

$$\kappa^2 = k_5 \dots k_8 \oplus k_9 \dots k_{12}$$

$$\kappa^3 = k_9 \dots k_{12} \oplus k_1 \dots k_4$$

- The round key function $f_{\kappa^i}(R)$ is simply obtained by XORing an a 4-bit input string R with the round key κ^i .

Exercise



- With a 12-bit system key $\kappa = ABC$ (represented in hex form), use this Feistel cryptosystem to encrypt the 8-bit plaintext $P = DF$ (represented in hex form)
- Perform the corresponding decryption to the ciphertext that resulted that resulted from P .

Data Encryption Standard (DES)

- In 1973 the US government initiated a competition to create an efficient cryptographic protocol that would be suitably secure for financial transactions and business communications to serve as a national standard.
- The public announcement was made by the National Bureau of Standards (NBS) which is now known as the National Institute of Standards and Technology (NIST).
- The outcome of this competition was the selection of the cipher Lucifer designed by IBM scientist lead by Horst Feistel (1915-1990).
- This cipher is called DES or Data Encryption Standard.

Data Encryption Standard (DES)



- The detailed description of DES is out of scope for these lectures.
- DES is a Feistel cryptosystem.
- The general structure of Feistel Cryptosystems is described in the previous slides.

Substitution-Permutation Network (SPN)

- A plaintext and ciphertext both are binary vectors of length ℓm , i.e., they are elements of $\{0, 1\}^{\ell m}$. The integer ℓm is said to be the block-length of the cipher. Two components of an SPN are
 - A substitution function that is also known as an S-box:
$$\pi_S: \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$$
 - A permutation function: $\pi_P: [\ell m] \rightarrow [\ell m]$ where $[\ell m] = \{1, \dots, \ell m\}$.

Substitution-Permutation Network

- Let ℓ , m , and NR be positive integers, let $\pi_S: \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ be a permutation, and $\pi_P: [\ell m] \rightarrow [\ell m]$ be a permutation. Let $\mathcal{P} = \mathcal{C} = \{0, 1\}^{\ell m}$.
- Let $\mathcal{K} \subseteq (\{0, 1\}^{\ell m})^{NR+1}$ consists of all possible key schedules that could be derived from an initial key K using the key scheduling algorithm. For a key schedule $(K^1, \dots, K^{NR+1})$, we encrypt the plaintext x using the algorithm in the next slide.

Substitution-Permutation Network

- The plaintext $x = x_{\langle 1 \rangle} || x_{\langle 2 \rangle} || \cdots || x_{\langle m \rangle}$
where $x_{\langle i \rangle} \in \{0,1\}^\ell$.
- The r th round input is $u^r =$
 $u_{\langle 1 \rangle}^r || u_{\langle 2 \rangle}^r || \cdots || u_{\langle m \rangle}^r$ where $u_{\langle i \rangle}^r \in \{0,1\}^\ell$.
- Substitution output $v^r =$
 $v_{\langle 1 \rangle}^r || v_{\langle 2 \rangle}^r || \cdots || v_{\langle m \rangle}^r$ where $v_{\langle i \rangle}^r \in \{0,1\}^\ell$.
- Permutation output $w^r =$
 $w_{\langle 1 \rangle}^r || w_{\langle 2 \rangle}^r || \cdots || w_{\langle m \rangle}^r$ where $w_{\langle i \rangle}^r \in \{0,1\}^\ell$
- The ciphertext $y = y_{\langle 1 \rangle} || y_{\langle 2 \rangle} || \cdots || y_{\langle m \rangle}$
where $y_{\langle i \rangle} \in \{0,1\}^\ell$.

Algorithm: Substitution-Permutation Network



Algorithm: Substitution-Permutation Network

Input : $x, \pi_S, \pi_P, (K^1, \dots, K^{NR+1})$

Output : y

1 : $w^0 \leftarrow x$

2 : for $r = 1$ to $NR - 1$

3 : $u^r \leftarrow w^{r-1} \oplus K^r$

4 : for $i = 1$ to m

5 : $v_{\langle i \rangle}^r \leftarrow \pi_S(u_{\langle i \rangle}^r)$

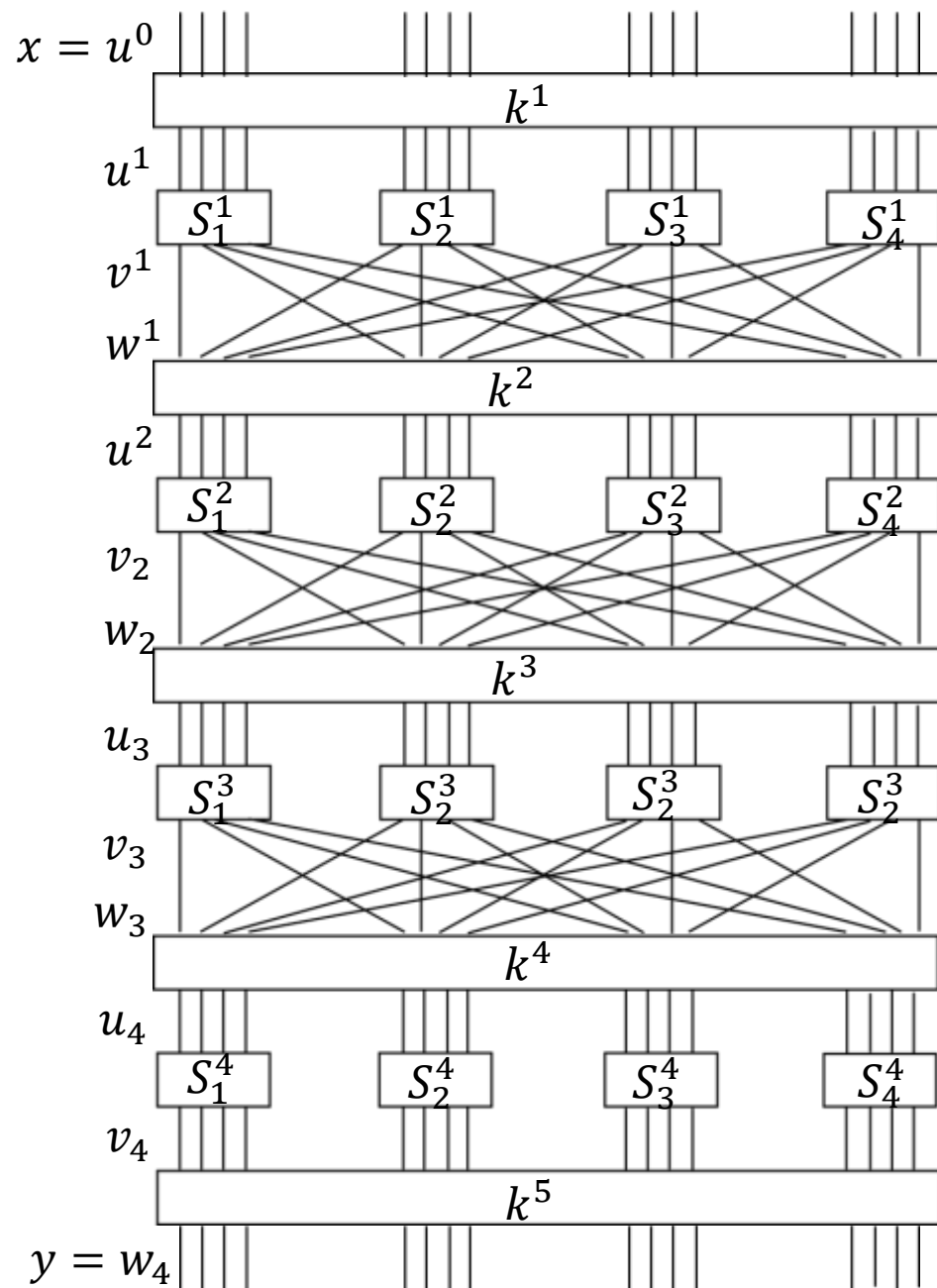
6 : $w^r \leftarrow (v_{\langle \pi_P(1) \rangle}^r, \dots, v_{\langle \pi_P(\ell m) \rangle}^r)$

7 : $u^{NR} \leftarrow w^{NR-1} \oplus K^{NR}$

8 : for $i = 1$ to m

9 : $v_{\langle i \rangle}^{NR} \leftarrow \pi_S(u_{\langle i \rangle}^{NR})$

10 : $y \leftarrow v^{NR} \oplus K^{NR+1}$



End of Week 3

Module 1

- Symmetric Ciphers
- Block ciphers
- SPN
- Feistel Structure referring to DES

Week 3

Module 2

- Modes of operation of a block cipher

Modes of operation of a block cipher

- Electronic Codebook (ECB) mode
- Cipher Block Chaining (CBC) mode
- Cipher Feedback (CFB) mode
- Output Feedback (OFB) mode

Modes of operations for block ciphers

Notation:

- We denote a sequence of plaintext blocks by $P_1, P_2, P_3, \dots$; each will be a block of ℓ bits (ℓ = the block length).
- The encryption mapping of the particular block cipher being employed is denoted as E_K , and its inverse (the decryption mapping) is denoted as $D_K \triangleq E_K^{-1}$.
- The corresponding blocks of ciphertexts that the modes of operation will send are denoted as $C_1, C_2, C_3, \dots$, each of length ℓ .

Electronic Codebook (ECB)

- Encrypt consecutive blocks (with the same key) and transmit the corresponding ciphertext blocks.
- This is called the electronic codebook (ECB) mode. This is represented by the equation
$$C_i = E_{\kappa}(P_i), \quad i = 1, 2, 3, \dots$$
where P_i is the plaintext, C_i is the ciphertext, and κ is the key.
- To decrypt, the recipient need only apply the inverse encryption mapping to the ciphertext blocks: $P_i = D_{\kappa}(C_i)$.

Electronic Codebook (ECB) Mode

- ECB mode is susceptible to a codebook attack.
- In a codebook attack:
 - The adversary collects and analyzes large sets of intercepted cipher blocks with whatever tools and additional information might be available, such as frequency analysis, cribs (known plaintext/ciphertext pairs).
 - Once certain cipher-block is ascertained, they are entered into a codebook that can be used to check subsequent messages for partial decryptions without actually having the key.
- To avoid such attacks, it is best to use a mode that adds some additional noise in the processed blocks so that identical plaintext blocks need not be processed into identical cipher blocks.

Cipher Block Chaining (CBC) Mode

- The cipher-block chaining (CBC) mode first XORs each plaintext block P_i with the previously produced ciphertext block C_{i-1} .
- The equation for CBD mode is
$$C_i = E_K(P_i \oplus C_{i-1}), \quad i = 1, 2, 3, \dots$$
- The zeroth ciphertext block C_0 is missing in this scheme so far.
- If we use the same C_0 for repeated transmission, then the corresponding first ciphertext blocks C_1 would be susceptible to a codebook attack.
- The zeroth ciphertext block is randomly produced and sent to the receiver without encryption at the time of the start of each new transmission.
- Decryption
$$P_i = D_K(C_i) \oplus C_{i-1}, \quad i = 1, 2, 3, \dots$$

Example

- Consider the following block encryption function $E_\kappa = E$ on 2-bit blocks (so the block size is $\ell = 2$) that is defined as follows:

Block Encryption function				
P	00	01	10	11
$E(P)$	10	00	11	01

For the plaintext: 1010100011

- Determine the corresponding ciphertext sequence that gets transmitted if electronic codebook (ECB) mode is used.
- Determine the corresponding ciphertext sequence that gets transmitted if cipher-block chaining mode is used with seed $C_0 = 10$.

Cipher Feedback (CFB) mode

- ECB and CBC modes process entire ℓ -bit blocks at a time, the CFB mode works on smaller subblocks of any size k , where $k \mid \ell$.
- A typical value of $k = 8$, so that each subblock represent one of the 256 ASCII characters.
- Since the CFB mode operates on subblocks of size k , each plaintext block P_i is decomposed into its $n = \frac{\ell}{k}$ subblocks:
$$P_i = p_i^1 p_i^2 \cdots p_i^n, \quad i = 1, 2, 3, \dots$$
- Thus each p_j^m represents a single k -bit subblock.

Cipher Feedback (CFB) Mode Encryption



Assume that we have a string of a certain number of plaintext subblocks $p_1, p_2, p_3, \dots$, that we need to encrypt and send. The resulting stream of cipher subblocks that the algorithm produces will be denoted as $c_1, c_2, c_3, \dots$. The encryption mapping E_K works with strings of n k -bit subblocks.

Cipher Feedback (CFB) Mode Encryption



Cipher Feedback (CFB) Mode Encryption

- Step 1. Generate an ℓ -bit shift register S_1 . Random generation is recommended. Initialize subblock counter $i = 1$.
- Step 2. Encrypt $S_i \rightarrow E_\kappa(S_i)$, let T_i be the leftmost k -bit subblock of $E_\kappa(S_i)$, and let R_i be the string of the rightmost $n - 1$ subblocks of the shift register S_i .
- Step 3. Set $c_i = p_i \oplus T_i$, update the next shift register as $S_{i+1} = R_i || c_i$, and update $i \rightarrow i + 1$.
- Step 4. Return to Step 2 until all plaintext subblocks have been encrypted.
-

Example: Suppose that we are given a plaintext 101110. Determine the ciphertext that gets transmitted if the encryption function is as given below.

Block Encryption function				
P	00	01	10	11
$E(P)$	10	00	11	01

Cipher Feedback (CFB) Mode Decryption



Cipher Feedback (CFB) Mode Decryption

- Step 1. Initialize subblock counter $i = 1$.
 - Step 2. Encrypt $S_i \rightarrow E_K(S_i)$, let T_i be the leftmost k -bit subblock of $E_K(S_i)$, and let R_i be the string of the rightmost $n - 1$ subblocks of the shift register S_i .
 - Step 3. Define $p_i = c_i \oplus T_i$, update the next shift register as $S_{i+1} = R_i || c_i$, and update $i \rightarrow i + 1$.
 - Step 4. Return to Step 2 until all ciphertext subblocks have been decrypted.
-

Output Feedback (OFB) Mode



- The CFB mode is a stream cipher protocol that can be used with any block cipher.
- If any error were to be introduced in transmitting any ciphertext subblock, the error would propagate through the stream to corrupt the next $n - 1$ decrypted characters, after which the corrupted stream would be flushed from the system.
- Output Feedback (OFB) mode is a more robust algorithm.

Out Feedback (OFB) Mode Encryption



Output Feedback (OFB) Mode Encryption

- Step 1. Generate an ℓ -bit shift register S_1 . Random generation is recommended. Initialize subblock counter $i = 1$.
- Step 2. Encrypt $S_i \rightarrow E_\kappa(S_i)$, let T_i be the leftmost k -bit subblock of $E_\kappa(S_i)$, and let R_i be the string of the rightmost $n - 1$ subblocks of the shift register S_i .
- Step 3. Define $c_i = p_i \oplus T_i$, update the next shift register as $S_{i+1} = R_i || T_i$, and update $i \rightarrow i + 1$.
- Step 4. Return to Step 2 until all plaintext subblocks have been encrypted.
-

End of Week 3

Module 2

- Modes of operation:
 - Electronic Codebook (ECB)
 - Cipher-block Chaining (CBC)
 - Cipher Feedback (CFB)
 - Output Feedback (OFB)

Week 3

Module 3

- Attack models.
- Linear approximation attack.

Cryptanalytic Attack Model

The following are the cryptanalytic attack model:

- Ciphertext only attack
- Known plaintext attack
- Chosen plaintext attack
- Chosen ciphertext attack

Ciphertext only attack



- The adversary possesses a string of ciphertext.
- Example: the attacks on classical ciphers.

Known plaintext attack



- The adversary possesses a string of plaintext, $\mathbf{x}$, and the corresponding ciphertext $\mathbf{y}$.
- In the process, several plaintext-ciphertext pairs are known to the adversary.
- The goal of the adversary is to find the unknown keys used by the legitimate intended participants of the communication network.

Chosen plaintext attack



- The adversary has the capacity of choosing any plaintext and get it encrypted by the sender.
- In other word, the adversary has at least temporary access to the encrypting device. Hence, he can choose a plaintext string, $\mathbf{x}$, and construct the corresponding ciphertext string $\mathbf{y}$.

Chosen Ciphertext Attack



- The adversary has the capacity of choosing any ciphertext and get it decrypted by the receiver.
- In other word, the adversary has at least temporary access to the decrypting device. However, the key remains unknown to the adversary.

Examples



- Linear approximation attack was proposed by Matsui in 1993 to cryptanalyze DES. This is a known plaintext attack.
- Differential attack was proposed by Dinur and Shamir in 1991. This is a chosen ciphertext attack developed to attack DES.
- Over the last three decades, these attacks have evolved to attack stream ciphers.

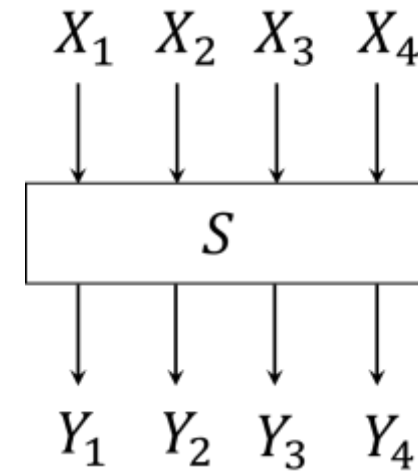
Week 3

Module 4

- Linear attack on SPN

Probabilistic Formulation

- In the probabilistic model of a cryptosystem:
 - The input variables and output variables are considered as random variables defined on $\{0, 1\}$.
 - $\Pr[X_i = 0] = p_i$ where $0 \leq p_i \leq 1$ is said to be the probability distribution of X_i .
 - $\epsilon_i = p_i - \frac{1}{2}$ is said to be the bias of X_i ; $-\frac{1}{2} \leq \epsilon_i \leq \frac{1}{2}$.
- The adjacent figure demonstrates this model for a single S-box.



Independence of random variables

- In general, let $X = (X_1, \dots, X_n)$, $Y = (Y_1, \dots, Y_m)$ be random variables corresponding to the input and output of an $n \times m$ S-box.
- It is reasonable to assume that the random variables $X_1, \dots, X_n$, are independent random variables.
- X_i, X_j are independent then
 - $\Pr[X_i = 0, X_j = 0] = p_i p_j$
 - $\Pr[X_i = 0, X_j = 1] = p_i(1 - p_j)$
 - $\Pr[X_i = 1, X_j = 0] = (1 - p_i)p_j$
 - $\Pr[X_i = 1, X_j = 1] = (1 - p_i)(1 - p_j)$.
- The distribution of the discrete random variable $X_i \oplus X_j$ is:
 - $\Pr[X_i \oplus X_j = 0] = p_i p_j + (1 - p_i)(1 - p_j)$
 - $\Pr[X_i \oplus X_j = 1] = p_i(1 - p_j) + (1 - p_i)p_j$.

Piling-up Lemma

Let $\epsilon_{\{i_1, \dots, i_k\}}$ denote the bias of the random variable $X_{i_1} \oplus \dots \oplus X_{i_k}$. Then $\epsilon_{i_1, \dots, i_k} = 2^{k-1} \prod_{j=1}^k \epsilon_{i_j}$ where ϵ_{i_j} is the bias of the random variable X_{i_j} , for $j = 1, \dots, k$.

- $$\begin{aligned} \Pr[X_{i_1} \oplus X_{i_2} = 0] &= p_{i_1} p_{i_2} + (1 - p_{i_1})(1 - p_{i_2}) \\ &= \left(\frac{1}{2} + \epsilon_{i_1}\right) \left(\frac{1}{2} + \epsilon_{i_2}\right) + \left(\frac{1}{2} - \epsilon_{i_1}\right) \left(\frac{1}{2} - \epsilon_{i_2}\right) = \frac{1}{2} + 2\epsilon_{i_1}\epsilon_{i_2} \end{aligned}$$
- $$\begin{aligned} \Pr[X_{i_1} \oplus X_{i_2} \oplus X_{i_3} = 0] &= \Pr[(X_{i_1} \oplus X_{i_2}) \oplus X_{i_3}] \\ &= \left(\frac{1}{2} + 2\epsilon_{i_1}\epsilon_{i_2}\right) \left(\frac{1}{2} + \epsilon_{i_3}\right) + \left(\frac{1}{2} - 2\epsilon_{i_1}\epsilon_{i_2}\right) \left(\frac{1}{2} - \epsilon_{i_3}\right) = \frac{1}{2} + 2^2\epsilon_{i_1}\epsilon_{i_2}\epsilon_{i_3} \end{aligned}$$

S -box Analysis: Linear Approximation

- Suppose that $S: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is an $n \times m$ S -box, or equivalently an (n, m) -function.
- Let $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_m)$ denote sequences of random variables that correspond to input and output, respectively.
- For each pair $(a, b) \in \mathbb{F}_2^n \times \mathbb{F}_2^m$ construct the following sum
$$a \cdot X \oplus b \cdot Y = \left(\bigoplus_{i=1}^n a_i X_i \right) \oplus \left(\bigoplus_{i=1}^m b_i Y_i \right).$$

S -box Analysis: Linear Approximation table

- Suppose $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$.
- $N_L(a, b) = \#\{ (x, y) : y = \pi_S(x), a \cdot x \oplus b \cdot y = 0 \}$
- For a 4×4 S -box

$$N_L(a, b) = \#\{ (x, y) : y = \pi_{S(x)}, (\bigoplus_{i=1}^4 a_i x_i) \oplus (\bigoplus_{i=1}^4 b_i y_i) = 0 \}$$

S -box Analysis: Linear Approximation table

- Suppose $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$.
- $N_L(a, b) = \#\{ (x, y): y = \pi_{S(x)}, a \cdot x \oplus b \cdot y = 0 \}$.
- For a 4×4 S -box
$$N_L(a, b) = \#\{ (x, y): y = \pi_{S(x)}, (\bigoplus_{i=1}^4 a_i x_i) \oplus (\bigoplus_{i=1}^4 b_i y_i = 0) \}$$



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S-box Analysis: Linear Approximation Table

x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$\pi_S(x)$	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7

a/b	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	16	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
1	8	8	6	6	8	8	6	14	10	10	8	8	10	10	8	8
2	8	8	6	6	8	8	6	6	8	8	10	10	8	8	2	10
3	8	8	8	8	8	8	8	8	10	2	6	6	10	10	6	6
4	8	10	8	6	6	4	6	8	8	6	8	10	10	4	10	8
5	8	6	6	8	6	8	12	10	6	8	4	10	8	6	6	8
6	8	10	6	12	10	8	8	10	8	6	10	12	6	8	8	6
7	8	6	8	10	10	4	10	8	6	8	10	8	12	10	8	10
8	8	8	8	8	8	8	8	8	6	10	10	6	10	6	6	2
9	8	8	6	6	8	8	6	6	4	8	6	10	8	12	10	6
A	8	12	6	10	4	8	10	6	10	10	8	8	10	10	8	8
B	8	12	8	4	12	8	12	8	8	8	8	8	8	8	8	8
C	8	6	12	6	6	8	10	8	10	8	10	12	8	10	8	6
D	8	10	10	8	6	12	8	10	4	6	10	8	10	8	8	10
E	8	10	10	8	6	4	8	10	6	8	8	6	4	10	6	8
F	8	6	4	6	6	8	10	8	8	6	12	6	6	8	10	8

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111

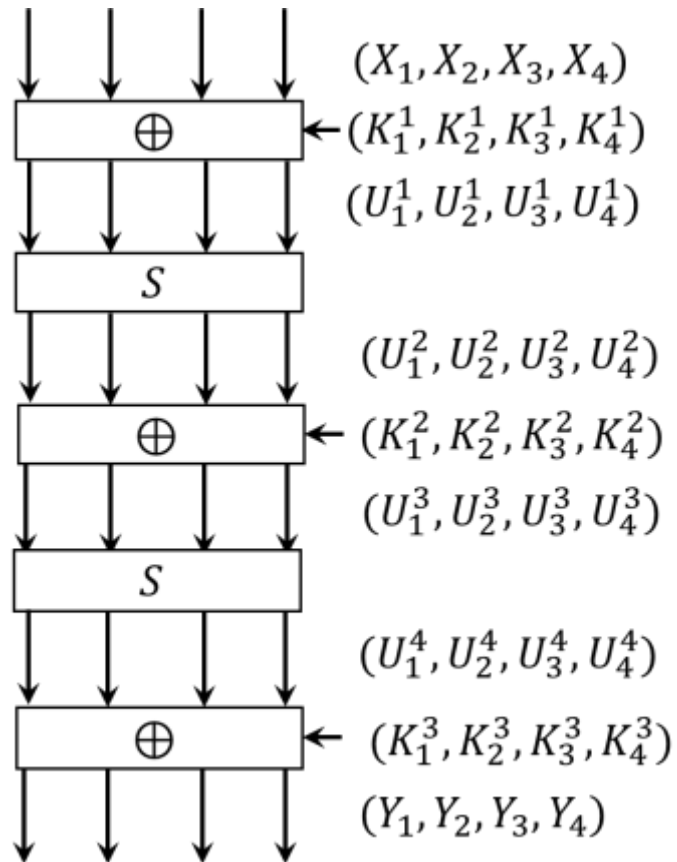
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	0111

Week 3

Module 5

- Linear attack on SPN

A simple block cipher

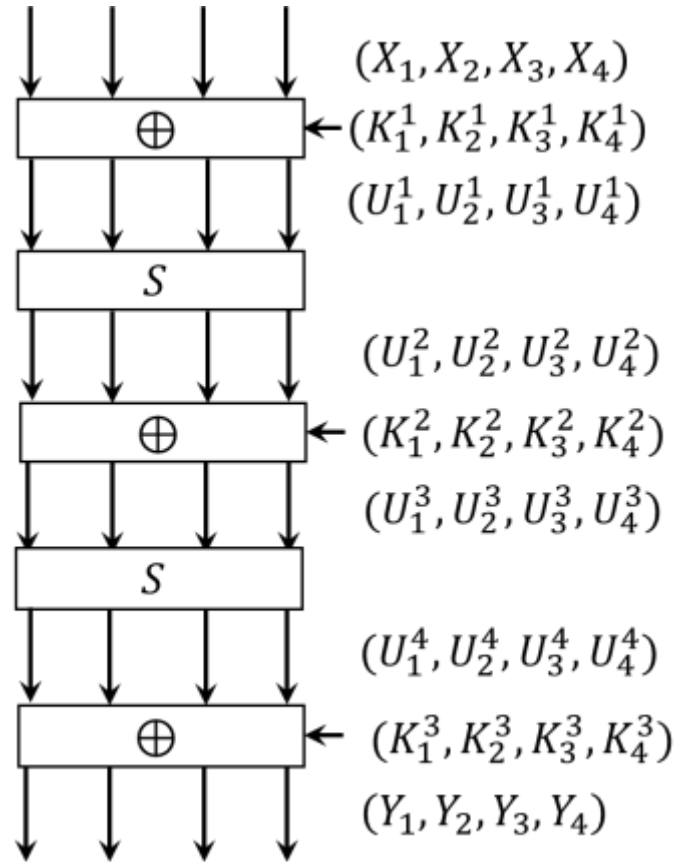


- $N_L(a, b) = \# \{ (x, y): y = \pi_S(x), (\oplus_{i=1}^4 a_i x_i) \oplus (\oplus_{i=1}^4 b_i y_i) = 0 \}$
- $\epsilon(a, b) = \frac{N_L(a, b) - 8}{16}$.
- From the linear approximation table, we observe:
 $N_L(1, 2) = N_L(1, 3) = 6$
 $N_L(6, 3) = N_L(A, 1) = N_L(B, 1) = N_L(B, 4) = N_L(C, 2) = 12$
- The biases are: $\epsilon(1, 2) = \epsilon(1, 3) = -\frac{1}{8}$, $\epsilon(6, 3) = \epsilon(A, 1) = \epsilon(B, 1) = \epsilon(B, 4) = \epsilon(C, 2) = \frac{1}{4}$.
- $N_L(A, 1) = 12$

Demonstration of the linear attack

- $\Pr[(X_2 \oplus X_4 \oplus Y_1 \oplus K_1^2) = 0] \in \left\{ \frac{12}{16}, \frac{4}{16} \right\}. (N_L(A, 1) = 12)$
- Steps of the linear attack:
 - Suppose that we have a sample of plaintext-ciphertext pairs, $(x_1^j, x_2^j, x_3^j, x_4^j), (y_1^j, y_2^j, y_3^j, y_4^j)$, for $j = 1, \dots, T$.
 - We assume a value of K_5 , say k_5 and compute
 - $S = x_2^j \oplus x_4^j \oplus y_1^j \oplus k_4^2$, for $j = 1, \dots, T$.
 - The theory tells us that if the guess of x_5 is correct then the number of times $S = x_2^j \oplus x_4^j \oplus y_1^j \oplus k_1^2$ divided by T is approximately $\frac{12}{16}$ or $\frac{4}{16}$.

A slightly more complicated cipher



- $N_L(A, 1) = 12, N_L(1, 7) = 14$.
- $T_1 = U_2^1 \oplus U_4^1 \oplus U_1^2$ has bias $\frac{1}{4}$;
- $T_2 = U_1^3 \oplus U_1^4 \oplus U_2^4 \oplus U_3^4$ has bias $\frac{3}{8}$.
- Since $U_1^3 = U_1^2 \oplus K_1^2$,
- $T_1 \oplus T_2 = U_2^1 \oplus U_4^1 \oplus K_1^2 \oplus U_1^4 \oplus U_2^4 \oplus U_3^4$.
- In the next slide we consider $T_1 \oplus T_2$.

$$T_1 \oplus T_2$$

$$\begin{aligned} \bullet \quad T_1 &= U_2^1 \oplus U_4^1 \oplus U_1^2; & U_1^3 &= U_1^3 \oplus K_1^2; \\ T_2 &= U_1^3 \oplus U_1^4 \oplus U_2^4 \oplus U_3^4 \\ T_1 &= U_2^1 \oplus U_4^1 \oplus U_1^2; & T_2 &= U_1^2 \oplus K_1^2 \oplus U_1^4 \oplus U_2^4 \oplus U_3^4 \end{aligned}$$

$$\begin{aligned} \bullet \quad T_1 \oplus T_2 &= U_2^1 \oplus U_4^1 \oplus K_1^2 \oplus U_1^4 \oplus U_2^4 \oplus U_3^4 \\ &= X_2 \oplus K_2^1 \oplus X_4 \oplus K_4^1 \oplus K_1^2 \\ &\quad \oplus Y_1 \oplus K_1^3 \oplus Y_2 \oplus K_2^3 \oplus Y_3 \oplus K_3^3 \\ &= X_2 \oplus X_4 \oplus Y_1 \oplus Y_2 \oplus Y_3 \oplus K_1^3 \\ &\quad \oplus K_2^3 \oplus K_3^3 \oplus K_2^1 \oplus K_4^1 \oplus K_1^2. \end{aligned}$$

$$T_1 \oplus T_2$$

- $\text{bias}(T_1) = \frac{1}{4}, \text{bias}(T_2) = \frac{3}{8}.$

- By Piling-up lemma

$$\begin{aligned} & \text{bias}(T_1 \oplus T_2) \\ &= 2 \times \frac{1}{4} \times \frac{3}{8} = \frac{3}{16}. \end{aligned}$$

Steps of Linear Attack

- Suppose that we have a sample of plaintext-ciphertext pairs, $(x_1^j, x_2^j, x_3^j, x_4^j), (y_1^j, y_2^j, y_3^j, y_4^j)$, for $j = 1, \dots, T$.
- We assume a value of k_1^3, k_2^3, k_3^3 , and compute
$$= x_2^j x_4^j \oplus y_1^j \oplus y_2^j \oplus y_3^j \oplus k_1^3 \oplus k_2^3 \oplus k_3^3, \text{ for } j = 1, \dots, T.$$
- The theory says that if the guess of x_5 is correct then the number of times $S = x_2^j \oplus x_4^j \oplus y_1^j \oplus k_5$ divided by T is approximately $\frac{11}{16}$ or $\frac{5}{16}$.